

**Chemistry, 2017 EC, entrance exam, version 4**

Grade 12 · Chemistry · 2017 E.C.

**Time allowed:** 120 minutes**Questions:** 72**Total marks:** 72

Name \_\_\_\_\_ Grade \_\_\_\_\_ ID No. \_\_\_\_\_ Date \_\_\_\_\_

1. Which of the following statements belongs to the limitation of Dalton's atomic theory? (1)
- A. An element is made up of smaller particle called atom
  - B. Atoms are made up of protons, neutrons and electrons
  - C. Atoms of the same element may have the same mass and properties
  - D. Atoms can neither be created nor destroyed during an ordinary chemical reaction
2. Which observation led J.J. Thomson to propose the 'plum pudding' model of the atom? (1)
- A. Measurement of electron charge using the oil drop experiment
  - B. Emission of X-rays by atoms
  - C. Deflection of alpha particles by gold foil
  - D. Discovery of electrons in cathode rays
3. According to modern atomic theory, which of the following subatomic particles is responsible for the atomic number of an element? (1)
- A. Protons
  - B. Nucleons
  - C. Neutrons
  - D. Electrons
4. Which of the following types of radioactive decays involves the emission of a particle consisting of two protons and two neutrons? (1)
- A. Gamma decay
  - B. Positron emission
  - C. Alpha decay
  - D. Beta decay
5. 'The atom has a small, dense, positively charged nucleus around which electrons rotate at high speed and relatively long distance from it'. This postulate is according to which model of the atom? (1)
- A. Rutherford's
  - B. Thomson's
  - C. Bohr's
  - D. Dalton's
6. Unknown element X has a valence shell electron configuration of  $3s^23p^5$ . Which of the following is true about element X? (1)
- A. It is stable and belongs to period 7
  - B. It forms a compound with others by losing electrons
  - C. It belongs to a class of noble gases
  - D. It belongs to group VIIA and period 3
7. Which properties of elements increase across a period in the periodic table? (1)
- A. Number of shells
  - B. Electronegativity
  - C. Metallic character
  - D. Atomic size

8. Which one of the following electromagnetic radiations has the highest frequency? (1)

- A. UV rays  
B. Microwaves  
C. X-rays  
D. Gamma rays

9. Which of the following represents an acceptable or possible set of quantum numbers (in the order  $n, l, m_l, m_s$ ) for an electron in an atom? (1)

- A. 3,1,-1,+1/2      B. 3,0,2,-1/2  
C. 3,2,0,0      D. 3,0,2,+1/2

10. Which property of elements increases both as you move from left to right across a period and from bottom to top within a group? (1)

- A. Metallic character  
B. Nuclear charge  
C. Atomic radius  
D. Electronegativity

11. Which of these elements has the highest second ionization energy? (1)

- A. Magnesium (Mg)                      B. Silicon (Si)  
C. Sodium (Na)                          D. Aluminum (Al)

**12.** Which of the following statements is correct according to Ernest Rutherford's famous Gold Foil Experiment? (1)

- A. Most alpha particles passed straight through, with a few deflected at large angles.
- B. All alpha particles were absorbed by the foil.
- C. All alpha particles passed straight through the gold foil.
- D. Most alpha particles were deflected at large angles.

**13.** Which scientist is known for formulating the uncertainty principle, which states that the position and momentum (1) of a particle cannot both be known exactly at the same time?

- A.** Niels Bohr  
**B.** Erwin Schrödinger  
**C.** Albert Einstein  
**D.** Werner Heisenberg

**14.** In the quantum mechanical model, the azimuthal quantum number for an electron in a 4f orbital is (1)

- A. 3  
B. 4  
C. 1  
D. 5

15. Which quantum number determines the orientation of an orbital in space? (1)

- A. Azimuthal quantum number  
B. Spin quantum number  
C. Principal quantum number  
D. Magnetic quantum number

16. According to Hund's Rule, when electrons occupy orbitals of equal energy, they will (1)

- A. fill each orbital singly with parallel spins before pairing
- B. fill orbitals in random order regardless of spin
- C. pair up in each orbital first before occupying the next
- D. fill the lowest energy orbital available

17. Which of the following factors does NOT affect the kinetic energy of photoelectrons in the photoelectric effect? (1)

- A. Work function of the material  
B. Frequency of incident light  
C. Wavelength of the incident light  
D. Intensity of incident light

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18. Which of the following statements best explains why a red light has a lower frequency than a blue light? (1)
- A. Red light contains more photons than blue light.
  - B. Red light undergoes greater refraction in a vacuum.
  - C. Red light travels slower than blue light.
  - D. Red light has a longer wavelength than blue light.
- 
19. When comparing infrared (IR) and ultraviolet (UV) radiation, which of the following is true? (1)
- A. IR has a shorter wavelength and higher energy than UV radiation.
  - B. Both IR and UV have the same frequency but different energies.
  - C. IR has a higher frequency than UV radiation.
  - D. IR has a longer wavelength and lower frequency than UV radiation.
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20. How does the energy of a photon relate to its wavelength in the electromagnetic spectrum? (1)
- A. Energy decreases as wavelength increases.
  - B. Energy is highest in visible light and lowest in gamma rays.
  - C. Energy increases with increasing wavelength.
  - D. Energy is independent of wavelength.
- 
21. What is the wavelength of light (nm) that has a frequency  $4.62 \times 10^{14} \text{ s}^{-1}$ ? (1)
- A.  $1.39 \times 10^{23}$
  - B.  $3.06 \times 10^{-19}$
  - C.  $1.54 \times 10^{-3}$
  - D. 649
- 
22. The energy of a photon in joule that has a wavelength of 12.3 nm is \_\_\_\_\_ J. (1)
- A.  $1.62 \times 10^{-17}$
  - B.  $1.99 \times 10^{-25}$
  - C.  $2.72 \times 10^{-50}$
  - D.  $4.42 \times 10^{-23}$
- 
23. Which one of the following represents an impossible set of quantum numbers for an electron in an atom? (1)  
(arranged as n, l, ml, ms)
- A. 1,0,0,1/2
  - B. 3,3,3,1/2
  - C. 2,1,1,1/2
  - D. 5,4,3,1/2
- 
24. The  $n = 5$  to  $n = 3$  transition in the Bohr hydrogen atom corresponds to the \_\_\_\_\_ of a photon with a wavelength of \_\_\_\_\_ nm. (1)
- A. emission, 657
  - B. absorption, 1270
  - C. absorption, 657
  - D. emission, 1280
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25. If an electron transitions from the  $n = 4$  level to the  $n = 2$  level in a hydrogen atom, what is the wavelength of the photon emitted? (1)
- A. 434 nm
  - B. 410 nm
  - C. 487 nm
  - D. 656 nm
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26. Which of the following properties of an electron in an atom does the principal quantum number determine? (1)
- A. Energy of the electron and distance from the nucleus
  - B. Spin direction of the electron
  - C. Orientation of the orbital in space
  - D. Shape of the orbital

27. Which of the following elements has the highest first ionization energy? (1)

- A. Oxygen (O)                      B. Fluorine (F)  
C. Carbon (C)                     D. Nitrogen (N)

28. Which of the following ions has the largest radius? (1)

- A.  $\text{Ca}^{2+}$   
B.  $\text{S}^{2-}$   
C.  $\text{Cl}^-$   
D.  $\text{K}^+$

29. Which of the following is correct regarding the atomic radii of elements as you move across a period and down a group?

- A. Atomic radius remains constant across a period and down a group  
B. Atomic radius increases both across a period and down a group  
C. Atomic radius decreases across a period and increases down a group  
D. Atomic radius increases across a period and decreases down a group

**30.** In a certain experiment NaOH solution is added from a burette to dilute sulphuric acid in a beaker. Which of the following changes will occur in the beaker? (1)

- A. concentration of hydroxide ion decreases  
B. pH increases  
C. concentration of hydrogen ion increases  
D. pOH increases

31. A solution of NaOH has a concentration of 0.001 M. What is the pH of this solution? (1)

- A. 13  
B. 14  
C. 11  
D. 12

**32.** According to Lewis definition, a base is a (an) \_\_\_\_\_ (1)

- A. electron pair donor  
B. proton acceptor  
C. electron pair acceptor  
D. proton donor

33. Which of the following is true about salts formed from weak acid and weak bases? (1)

- A. the type of salt formed has acidic property  
B. no hydrolysis takes place at all  
C. only the cation of the salt undergo hydrolysis  
D. both the cation and the anion of the salt undergo hydrolysis

**34.** Which of the following species is NOT amphoteric? (1)

- A.  $\text{H}_2\text{O}$  B.  $\text{NH}_4^+$   
C.  $\text{H}_2\text{PO}_4^-$  D.  $\text{HSO}_4^-$

**35.** What is the initial pH of a 0.1 M HCl solution before any NaOH is added in a titration? (1)

- A. 3.0  
B. 4.0  
C. 1.0  
D. 2.0

**36.** During the titration of 0.1 M HCl with 0.1 M NaOH, what is the pH of the solution at the equivalence point? (1)

- A. Exactly 7  
B. Depends on the concentration of the NaOH solution  
C. Less than 7  
D. Greater than 7

**37.** According to the Arrhenius definition, which of the following compounds is considered an acid? (1)

- A.  $\text{NH}_3$   
B.  $\text{CH}_4$   
C.  $\text{NaOH}$   
D.  $\text{HNO}_3$

38. Which of the following solutions would have the highest pH? (1)

- A. 0.01 M NaOH  
B. 0.1 M acetic acid  
C. 0.01 M HCl  
D. 0.05 M  $\text{NH}_4\text{OH}$

39. According to the Lewis definition, which of the following is a Lewis acid? (1)

- A.  $\text{BF}_3$   
C.  $\text{NH}_3$
- B.  $\text{CH}_4$   
D.  $\text{OH}^-$

40. Which of the following statements is true regarding the three acid-base theories? (1)

- A.** All Lewis bases are Arrhenius bases.
- B.** All Brønsted-Lowry acids are Arrhenius acids.
- C.** The Arrhenius theory can be applied in non-aqueous solvents.
- D.** All Brønsted-Lowry acids are Lewis acids.

41. The pH of a buffer made by mixing 0.2 M  $\text{NH}_4\text{Cl}$  and 0.2 M  $\text{NH}_3$  ( $K_b$  for  $\text{NH}_3 = 1.8 \times 10^{-5}$ ) is closest to: (1)

- A. 7.26                      B. 6.26  
C. 9.26                      D. 8.26

**42.** Which one of the following pairs can't be mixed together to form a buffer solution? (1)

- A.**  $\text{CH}_3\text{COOH}$  and  $\text{CH}_3\text{COONa}$       **B.**  $\text{H}_3\text{PO}_4$  and  $\text{NaH}_2\text{PO}_4$   
**C.**  $\text{HNO}_3$  and  $\text{NH}_4\text{NO}_3$       **D.**  $\text{NH}_3$  and  $\text{NH}_4\text{Cl}$

**43.** If  $\text{CH}_3\text{COONa}$  is added to a solution of  $\text{CH}_3\text{COOH}$ , which of the following statements is correct? (1)

- A.** The concentration of hydrogen ion increases  
**B.** The concentration of  $\text{CH}_3\text{COOH}$  decreases  
**C.** The dissociation of  $\text{CH}_3\text{COOH}$  decreases  
**D.** The pH of a solution decreases

44. A one liter solution is prepared by mixing 0.1 mole HCl and 0.05 mole NaOH. What is the resulting pH of the solution? (1)

- A. 3.3  
B. 4.2  
C. 1.3  
D. 2.5

**45.** A 50 mL sample of 2.0 M HCl is diluted with water to a final volume of 250 mL. What is the concentration of HCl in the diluted solution? (1)

- A. 0.5 M                      B. 0.8 M  
C. 0.2 M                      D. 0.4 M

46. During the titration of known volume of strong acid with strong base, there is (1)

- A. a low pH at the beginning of a titration
- B. no pH change of solution
- C. a formation of acidic salt
- D. a sharp decreasing of pH around the equivalence point



**55.** During the titration of 0.1 M acetic acid with 0.1 M NaOH, what can be said about the pH at the equivalence point? (1)

- A. It is greater than 7 because acetate ions form a basic solution.
- B. It depends on the concentration of NaOH.
- C. It is less than 7 because acetic acid is weak.
- D. It is equal to 7 because NaOH is strong.

**56.** Which one of the following is a conjugate acid-base pair? (1)

- A.  $\text{NH}_4^+$  and  $\text{NH}_3$   
B.  $\text{NaF}$  and  $\text{F}^-$   
C.  $\text{H}_2\text{O}$  and  $\text{H}_2\text{O}_2$   
D.  $\text{HNO}_3$  and  $\text{HNO}_2$

**57.** In a titration of acetic acid with sodium hydroxide, what is the pH of the solution before the equivalence point? (1)

- A. pH > 7  
B. Cannot be determined  
C. pH = 7  
D. pH < 7

**58.** A buffer solution contains acetic acid and sodium acetate. If the concentration of acetic acid equals the concentration of sodium acetate, what is the pH of the buffer solution? (1)

- A. 5.05  
B. 5.23  
C. 4.57  
D. 4.74

59. If the following reaction is balanced in acidic media, what will be the coefficient of  $\text{Fe}^{2+}$  in the balanced reaction?  $\text{Fe}^{2+} + \text{Cr}_2\text{O}_7^{2-} \rightarrow \text{Fe}^{3+} + \text{Cr}^{3+}$  (1)

- A. 6  
B. 3  
C. 14  
D. 1

60. If the following chemical reaction is balanced using a change in oxidation number method, what would be the coefficient of  $\text{SO}_2$ ?  $\text{PbS(s)} + \text{O}_2\text{(g)} \rightarrow \text{PbO(s)} + \text{SO}_2\text{(g)}$  (1)

- A. 5  
B. 4  
C. 2  
D. 3

**61.** Which of the following pairs represents a conjugate acid-base pair in the reaction:  $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$ ? (1)

- A.**  $\text{NH}_4^+$  and  $\text{H}_2\text{O}$  **B.**  $\text{NH}_4^+$  and  $\text{OH}^-$   
**C.**  $\text{NH}_3$  and  $\text{OH}^-$  **D.**  $\text{NH}_3$  and  $\text{NH}_4^+$

62. Which of the following salts will create a basic solution when dissolved in water? (1)

- A. FeCl<sub>3</sub>  
B. KBr  
C. NH<sub>4</sub>Cl  
D. Na<sub>2</sub>CO<sub>3</sub>

**63.** If the pKa of lactic acid is 3.86, which of the following ratios of lactate to lactic acid would give a pH of 4.86? (1)

- A. 1:1  
B. 1:5  
C. 1:10  
D. 10:1

**64.** What is the oxidation number of sulfur in the compound  $\text{Na}_2\text{S}_2\text{O}_7$  (sodium thiosulfate? Actually  $\text{Na}_2\text{S}_2\text{O}_7$  is sodium pyrosulfate, not thiosulfate. Thiosulfate is  $\text{Na}_2\text{S}_2\text{O}_3$ . So careful: question says 'sodium thiosulfate' but formula given is  $\text{Na}_2\text{S}_2\text{O}_7$ . Likely they mean pyrosulfate. Oxidation number calculation: Na (+1 each), O (-2 each). Let S = x.  $2(+1) + 2x + 7(-2) = 0 \rightarrow 2 + 2x - 14 = 0 \rightarrow 2x = 12 \rightarrow x = +6$ . (1)

- A. +2  
B. +3  
C. +4  
D. +6

65. In the following reaction, identify the species being oxidized and the species acting as the oxidizing agent:  $\text{Cu} + \text{HNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO}_2 + \text{H}_2\text{O}$  (1)
- A. Cu is oxidized, and  $\text{NO}_2$  is the reducing agent.  
B.  $\text{HNO}_3$  is oxidized, and  $\text{NO}_2$  is the reducing agent.  
C. Cu is oxidized, and  $\text{HNO}_3$  is the oxidizing agent.  
D.  $\text{HNO}_3$  is oxidized, and Cu is the oxidizing agent.
- 
66. To balance the following redox reaction in a basic solution, how many hydroxide ions ( $\text{OH}^-$ ) are needed?  $\text{MnO}_4^- + \text{I}^- \rightarrow \text{MnO}_2 + \text{I}_2$  (1)
- A. 4  
B. 8  
C. 2  
D. 6
- 
67. Given the following  $K_a$  constants:  $K_a(\text{HF})=7.2 \times 10^{-4}$ ,  $K_a(\text{HNO}_2)=4.5 \times 10^{-4}$ ,  $K_a(\text{HCN})=6.5 \times 10^{-10}$ . Which of the following is the correct order of the substances in basic strength? (1)
- A.  $\text{NO}_2^- > \text{F}^- > \text{CN}^-$   
B.  $\text{NO}_2^- > \text{CN}^- > \text{F}^-$   
C.  $\text{CN}^- > \text{NO}_2^- > \text{F}^-$   
D.  $\text{F}^- > \text{NO}_2^- > \text{CN}^-$
- 
68. Which acid has the highest percentage ionization in a 0.1 M solution, assuming equal initial concentrations? (1)
- A. Acid with  $K_a 4.5 \times 10^{-4}$   
B. Acid with  $K_a 2.1 \times 10^{-3}$   
C. Acid with  $K_a 1.8 \times 10^{-5}$   
D. Acid with  $K_a 7.5 \times 10^{-6}$
- 
69. A solution contains 1.0 M of a weak acid (HA) and 0.1 M of its conjugate base ( $\text{A}^-$ ). If the  $\text{p}K_a$  of the acid is 5.0, what is the pH of the solution? (1)
- A. 3.0  
B. 6.0  
C. 4.0  
D. 5.0
- 
70. For a weak acid with a  $\text{p}K_a$  of 4.5, what ratio of conjugate base to acid is needed to prepare a buffer with a pH of 5.5? (1)
- A. 100:1  
B. 1:100  
C. 1:10  
D. 10:1
- 
71. Which salt would you expect to create a pH closest to neutral when dissolved in water? (1)
- A. NaCN  
B.  $\text{NH}_4\text{Cl}$   
C.  $\text{NaNO}_3$   
D.  $\text{NH}_4\text{NO}_3$
- 
72. A solution of KCN has a pH greater than 7. Which ion in this salt is responsible for the basic pH? (1)
- A.  $\text{H}^+$   
B.  $\text{OH}^-$   
C.  $\text{K}^+$   
D.  $\text{CN}^-$

— END OF EXAMINATION —